Domain Shift in Human Activity Recognition

A Quantitative Analysis of WISDM vs. UCI HAR

RQ1 Technical Analysis

Sensor Data Analytics / Machine Learning Systems

March 2026

Research Question: *What are the key sources of domain shift across the WISDM dataset and other HAR datasets?*
What metrics can be used to quantify the differences?

Outline

1. Motivation
2. Dataset Overview
3. Methodology
4. Results
5. Conclusions

Why Does Domain Shift Matter in HAR?

The deployment gap

- Models trained on dataset A fail on dataset B — even for the *same* activity classes
- Accuracy can drop from **95%** (within-dataset) to <**60%** (cross-dataset)
- Root cause is rarely understood at the signal level

Why WISDM and UCI HAR?

- Two of the **most widely cited** HAR benchmarks
- Nominally identical task: 6-class accelerometer HAR
- Yet they embody *fundamentally different* data-generating processes

The Core Question

*What exactly makes these datasets different — and how do we **measure** it?*

Key Claim

Domain shift in HAR is **multi-source** and **anisotropic**.
Scalar metrics (e.g. mean Acc Magnitude) hide the dominant shift axis.

Dataset Overview — WISDM vs. UCI HAR

Property	WISDM v1.1	UCI HAR
Subjects	36	30
Sampling rate	20 Hz	50 Hz
Sensor placement	**Front trouser pocket** (free-swinging)	**Waist-mounted** (fixed)
Sensors	Accelerometer only	Accelerometer + **Gyroscope**
Gravity separation	**None** (raw total_acc)	**Butterworth LPF** applied
Activities	Walk, **Jog**, Up, Down, Sit, Stand	Walk, Up, Down, Sit, Stand, **Lay**
Total windows*	43,347	8,355
Collection setting	**In the wild**	**Controlled lab**

* After alignment: 20 Hz, 51-sample (2.56 s) windows

WISDM: Walk 38.5% · **Jog 31.0%** · other 30.5%

UCI HAR: Sit 21.3% · Stand 22.8% · Walk 20.6% · other

Label Space — Shared Activities and Gaps

ID	Unified Label	WISDM	UCI
0	Walking	✓	✓
1	**Jogging**	✓	—
2	Upstairs	✓	✓
3	Downstairs	✓	✓
4	Sitting	✓	✓
5	Standing	✓	✓
—	**Laying**	—	✓

Activity Gap

- **Jogging** occupies **31%** of WISDM — no UCI counterpart
- **Laying** is UCI-only — dropped from analysis
- Irresolvable by any preprocessing

5 Shared Classes

Walk, Upstairs, Downstairs, Sitting, Standing

Methodology Pipeline

Raw Data WISDM+UCI → Physical Alignment (50→20 Hz) → Feature Extraction (Acc_{mag}, f_c) → KDE Fitting (per axis) → Distance Metrics (Wass/KL/MMD)

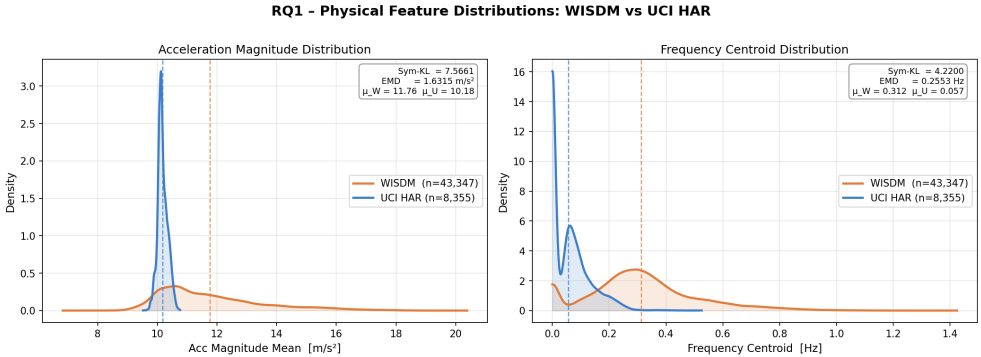
Two physics-motivated features per window:

- **Acc Magnitude:** $Acc_{mag}(t) = \sqrt{x^2 + y^2 + z^2}$
- **Frequency Centroid:** $f_c = \frac{\sum f_i \cdot P(f_i)}{\sum P(f_i)}$

Four distance metrics:

- **Wasserstein-1** — physical transport cost
- **Sym. KL Divergence** — distributional overlap
- **Multi-kernel MMD** — feature-space shift
- **Total Variation Distance** — prior $P(Y)$

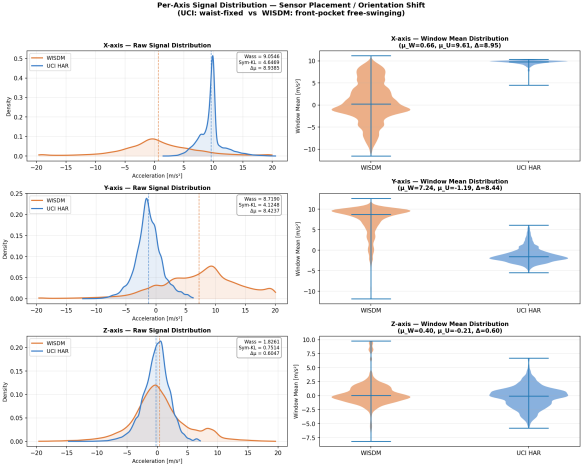
Results I — Physical Feature Distributions (Overview)



- WISDM: broad, right-skewed
- UCI: narrow spike near $1g = 9.81$ m/s²

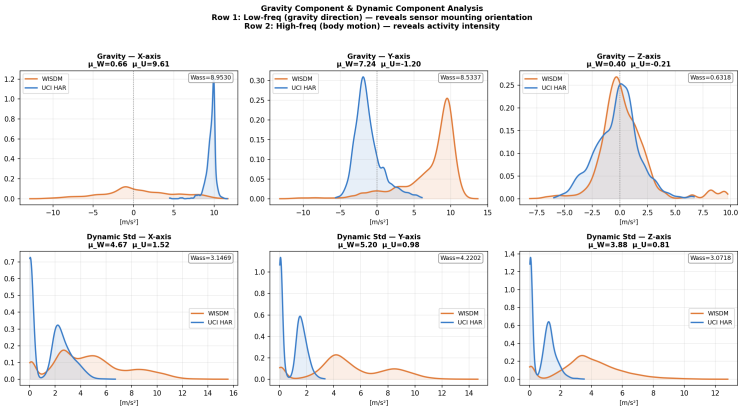
Frequency centroid gap: $\bar{f}_W = 0.312$ Hz vs. $\bar{f}_U = 0.057$ Hz **(5.5×)**

Results II — Dominant Source: Sensor Orientation



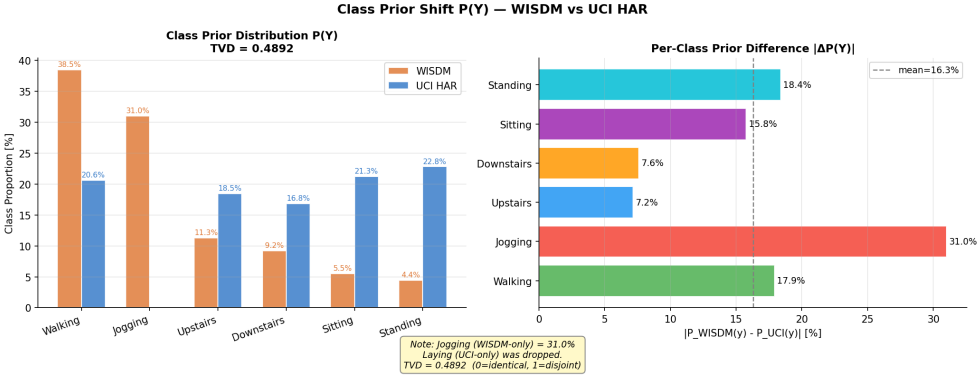
X-axis: Wass = 9.06 m/s² **Y-axis:** Wass = 8.72 m/s² **Z-axis:** Wass = 1.83 m/s² $\gg$ **Scalar Acc Mag = 1.63 m/s²**

Results III — Gravity Component & Preprocessing Gap



Row 1: Gravity on **X** in UCI (+8.58 m/s^2) vs. **Y** in WISDM (+6.51 m/s^2) — near-orthogonal sensor orientation **Row 2:** Dynamic std **3–5**$\times$ **higher** in WISDM (free pocket vs. fixed waist)

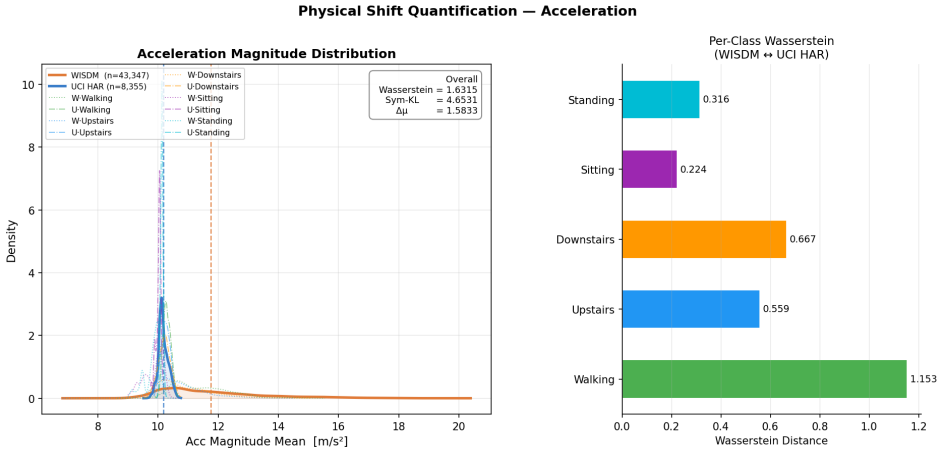
Results IV — Class Prior Shift $P(Y)$



Total Variation Distance $\mathbf{TVD} = \mathbf{0.489}$

Jogging $\Delta P = 31.0\%$ · Standing $\Delta P = 18.4\%$ · Walking $\Delta P = 17.9\%$

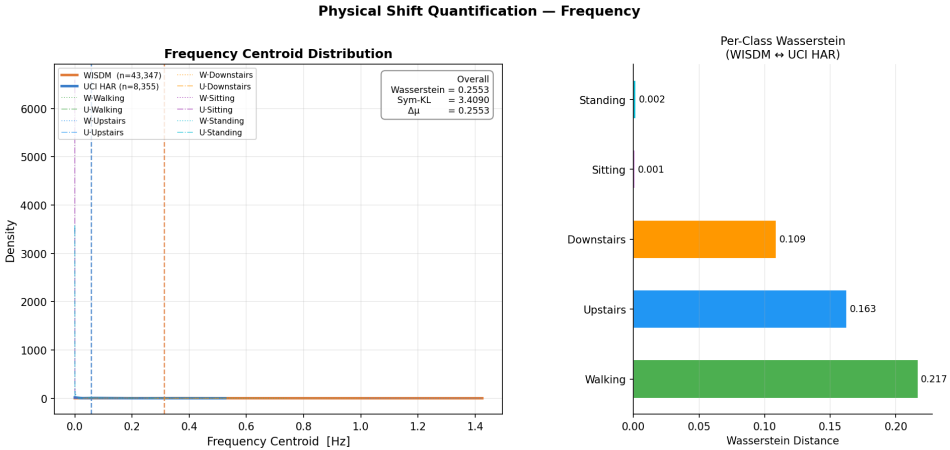
Results V-a — Per-Class Covariate Shift: Acc Magnitude



Left: KDE with per-class curves Right: per-class Wasserstein bar chart

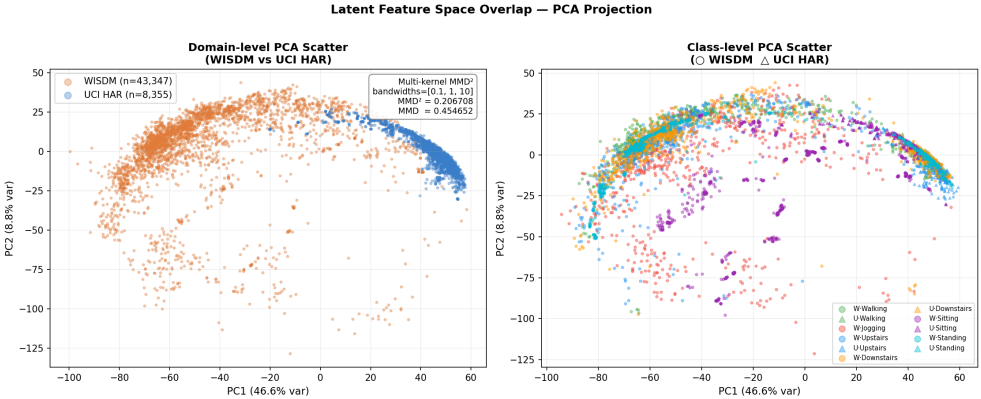
Walking dominates (Wass = 1.15 m/s^2) **Sitting / Standing** are nearly aligned across datasets

Results V-b — Per-Class Covariate Shift: Frequency Centroid



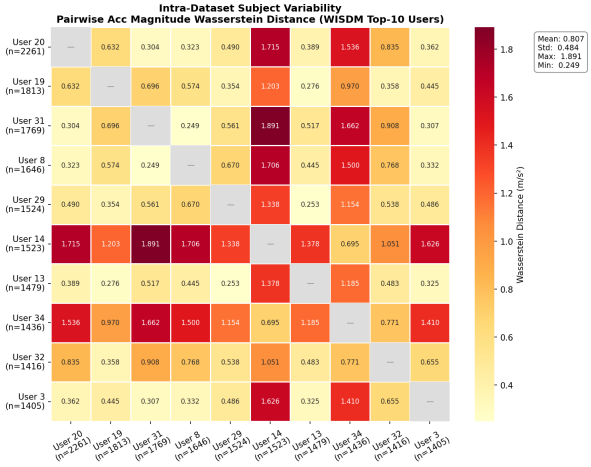
Dynamic activities (Walk 0.22 Hz, Upstairs 0.16 Hz) show significant frequency shift
Static activities (Sitting 0.001 Hz, Standing 0.002 Hz) are nearly **identical** across datasets

Results VI — Feature Space Geometry (MMD)



Multi-kernel MMD (bandwidths 0.1, 1, 10): $\mathbf{MMD^2 = 0.2067}$, $\mathbf{MMD = 0.455}$ Left: domain-coloured Right: class-coloured

Results VII — Inter-Subject Variability



Key Numbers

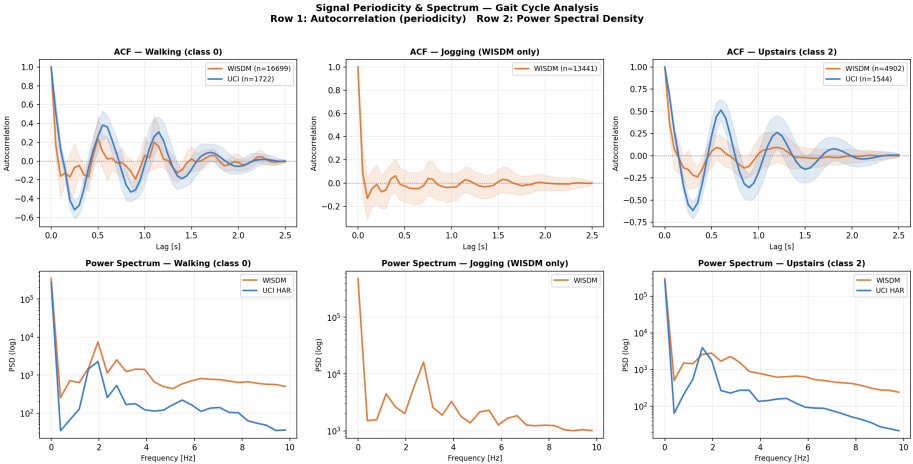
Mean intra-user Wass	0.807
Max intra-user Wass	**1.891**
Cross-dataset Wass	**1.631**

Critical Finding

Max user-pair distance **(1.89)** exceeds cross-dataset shift **(1.63)**.

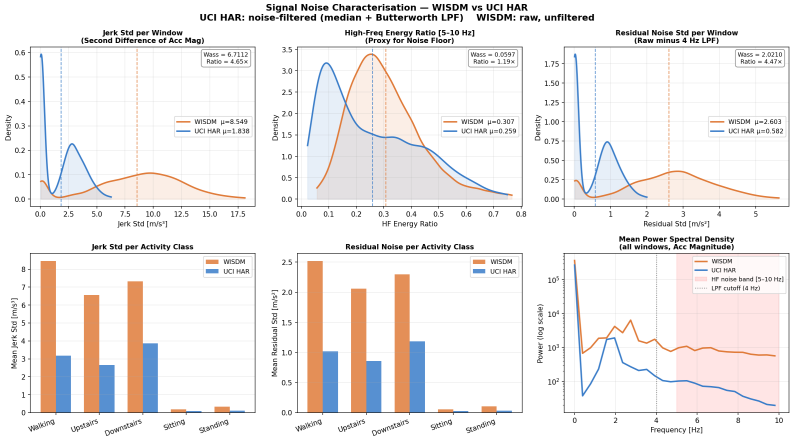
Subject shift is **not secondary** — it is a first-class problem.

Results VIII — Gait Periodicity & Spectral Analysis



Row 1: Autocorrelation — WISDM Walking shows stronger gait periodicity (pocket amplifies step motion)
Row 2: PSD — WISDM energy concentrated at 1–2 Hz step frequency

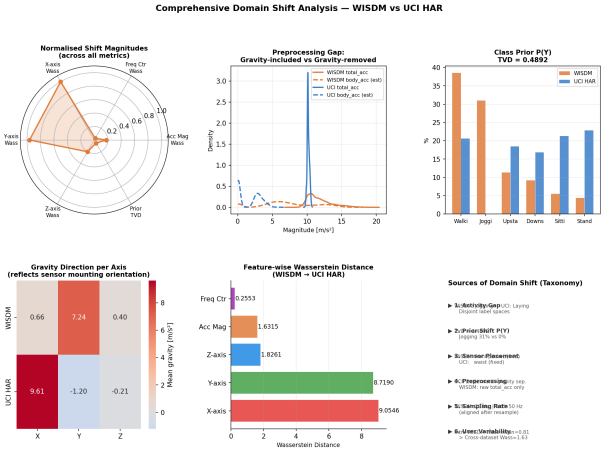
Results IX — Signal Noise Level



UCI HAR: median + Butterworth LPF applied before release **WISDM**: raw, unfiltered

Jerk Std **4.65**× higher in WISDM · Residual Noise Std **4.47**× higher · PSD diverges sharply above 4 Hz

Results Summary — Comprehensive View



Radar chart confirms: X/Y-axis Wasserstein dominates all other metrics after normalisation

Metric Summary Table

Feature / Dimension	Type	Wasserstein	Sym-KL	Note
X-axis (raw signal)	Covariate	**9.055 m/s²**	4.647	Dominant axis
Y-axis (raw signal)	Covariate	**8.719 m/s²**	4.125	Dominant axis
Z-axis (raw signal)	Covariate	1.826 m/s²	0.751	Stable
Acc Magnitude (global)	Covariate	1.631 m/s²	4.653	Understated
Acc Magnitude — Walking	Per-class	1.153 m/s²	5.617	Largest class shift
Frequency Centroid	Covariate	0.255 Hz	3.409	5.5× mean gap
Class Prior TVD	Prior	—	—	**0.489**
Multi-kernel MMD	Global	—	—	**0.455**
Max inter-user Wass	Subject	**1.891 m/s²**	—	> cross-dataset

Seven Sources of Domain Shift — Taxonomy

1. **Sensor placement / orientation**
   X/Y Wass $\approx$ 9 m/s^2; gravity projected onto different axes
2. **Preprocessing pipeline**
   UCI: Butterworth gravity removal; WISDM: raw total_acc
3. **Activity label space mismatch**
   Jogging (WISDM 31%) $\leftrightarrow$ Laying (UCI) — no cross-match
4. **Class prior shift** $P(Y)$
   TVD $= 0.489$
5. **Signal dynamic amplitude**
   WISDM dynamic std 3–5$\times$ higher across all axes
6. **Inter-subject variability**
   Max intra-dataset shift (1.89) $>$ cross-dataset (1.63)
7. **Signal noise level**
   Jerk 4.65$\times$, residual 4.47$\times$ higher in WISDM (unfiltered vs. Butterworth-filtered)
8. **Sensor modality gap**
   UCI has gyroscope; WISDM accelerometer only

Bottom Line

Sensor **orientation** is the dominant shift source — invisible to scalar analysis.

Recommended Metrics for Cross-Dataset HAR Shift

Metric	Captures	When to Use
Per-axis Wasserstein	Orientation shift	**Primary** — never use scalar alone
Scalar Wasserstein (Acc Mag)	Magnitude shift	Comparison with prior literature
Symmetric KL Divergence	Distributional overlap	Tail sensitivity, shape differences
Multi-kernel MMD	Global feature-space	Domain adaptation readiness
Total Variation Distance	Prior $P(Y)$ shift	Mandatory for imbalanced datasets
Gravity DC analysis	Orientation / preprocessing	Separates orientation from motion
Inter-user Wasserstein	Subject heterogeneity	Personalisation difficulty baseline

Practical Recommendation

Reporting scalar Acc Magnitude shift alone is **insufficient** and underestimates the true domain gap by $5\times$. A complete analysis requires **per-axis**, **prior**, **feature-space**, and **subject-level** metrics.

Conclusions

Answering the Research Question:

- ✓ **7 sources** of domain shift identified and quantified
- ✓ **Per-axis analysis** reveals the dominant shift missed by prior work
- ✓ **4 metric classes** cover covariate, prior, feature-space, and subject-level shift
- ✓ Intra-dataset user shift can **exceed** cross-dataset shift
- ✓ Preprocessing (gravity separation) is a **hidden covariate shift** — not just a technicality

Implications for DA Research

- Adaptation methods must address sensor **orientation** explicitly
- Standard gravity separation should be applied **consistently** across datasets
- Subject-level adaptation may matter **more** than dataset-level
- Evaluation should report **per-axis metrics**, not just Acc Mag

Repository

All code, data, and figures:
`github.com/gengzhanguo/wisdm_ucihar`

Thank You

Questions welcome

All figures, data, and reproducible code available at:

`https://github.com/gengzhanguo/wisdm_ucihar`

WISDM v1.1
Kwapisz et al., 2011

UCI HAR
Anguita et al., 2013